Empirical Bogomolny–Keating 2-point correction on 10^5 Odlyzko zeros, and a no-go for log-type saturation envelopes in complexity-bounded generalised second law inequalities

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We report two independent small-scale results arising from a pre-registered testing pipeline for the ECI v6 differential generalised-second-law inequality [1]. First: on the first 10^5 non-trivial Riemann zeros of Odlyzko's public tables, the empirical pair correlation $R_2^{\mathrm{emp}}(x)$ deviates from the pure Montgomery–Dyson GUE prediction at $\chi^2 = 434.19/58\,\mathrm{dof}$; the Bogomolny–Keating (1996) arithmetic 2-point correction accounts for this deviation with $\chi^2/\mathrm{dof} = 4.17$ at best-fit amplitude and effective height, leaving a residual $\nu \approx 1$ oscillation that is most plausibly an unfolding miscalibration rather than new structure. Second: in the Fan (2022) logarithmic Krylov scrambling regime, any saturation-envelope $\Phi(\mathcal{C}_k/\mathcal{C}_k^{\max})$ with $\Phi(1) \neq 0$ fails the boundary condition $dS_{\mathrm{gen}}/d\tau_R \to 0$; a recent log-type envelope proposal $\Phi(x) \propto \log(1 + x/x^*)$ is thus excluded, while the logistic $\Phi(x) = 1 - x$ of the v6.0.6 formal companion satisfies it. We report both as technical notes; neither modifies the companion paper [1].

I. CONTEXT

The v6.0.6 JHEP-formal companion [1] establishes a differential inequality

$$\frac{dS_{\mathrm{gen}}[R]}{d\tau_R} \;\leq\; \kappa_R\, \mathcal{C}_k[\rho_R(\tau_R)]\, \Theta(\mathrm{PH}_k[\delta n]), \tag{1}$$

with a tightened logistic envelope $\kappa_R\, \mathcal{C}_k(1 - \mathcal{C}_k/\mathcal{C}_k^{\max})\, \Theta$ at scrambling. During the pre-tag adversarial process, two adjacent questions arose that neither modify nor depend on (1) but admit short direct answers:

1. Does the pair-correlation of Riemann zeros support a specific beyond-GUE correction of the Connes–Consani–Moscovici 2025 spectral-triple form [6], or is the classical Bogomolny–Keating 1996 correction [2] sufficient?
2. Is the logistic shape of the v6.0.6 saturation envelope a specific choice, or does the Fan (2022) Krylov saturation condition constrain the admissible envelope class?

We report both results. All computations are executed against pre-registered decision thresholds; all raw artefacts are public.

II. EMPIRICAL 2-POINT CORRECTION ON 10^5 ODLYZKO ZEROS

A. Data and method

We download the first 10^5 imaginary ordinates γ_n of the non-trivial zeros of $\zeta(1/2 + is)$ from Odlyzko's public table [8]. The sample spans $\gamma \in [14.13,\ 74920.83]$.

Normalised spacings use the Riemann–von Mangoldt smooth term $\bar{\rho}(\gamma) = (1/2\pi)\log(\gamma/(2\pi))$:

$$\tilde{\gamma}_n = \frac{\gamma_n}{2\pi}\log\frac{\gamma_n}{2\pi} - \frac{\gamma_n}{2\pi} + \frac{7}{8}. \tag{2}$$

The unfolded spacings $\delta_n = \tilde{\gamma}_{n+1} - \tilde{\gamma}_n$ have mean 1.000000 and standard deviation 0.4009, consistent with GUE.

The empirical 2-point function $R_2^{\mathrm{emp}}(x)$ is estimated on a grid $x \in (0, 3]$, $\Delta x = 0.05$ (60 bins). Error bars are max(Poisson, 10-block jackknife).

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B. GUE baseline

Against the Montgomery–Dyson prediction $R_2^{\mathrm{GUE}}(x) = 1 - (\sin \pi x/\pi x)^2$, we obtain

$$\chi^2_{\mathrm{GUE}} \;=\; 434.19\,/\,58\,\mathrm{dof} \;=\; 7.49, \qquad p \to 0.$$

Peak residual at $x \approx 0.23$ reaches 7.74σ; the residual is systematically negative on $x \in [0.1, 0.7]$ and oscillates at higher x (Fig. 1). Pure GUE is ruled out at this height.

C. Bogomolny–Keating fit

We fit the Berry–Keating arithmetic correction in the unfolded-x convention [3, eq. (4.20)], which after substituting $\langle d \rangle = L/(2\pi)$ reads

$$\Delta_{\mathrm{BK}}(x; L) = -\frac{2}{L^2} \sum_p \frac{(\log p)^2}{(p-1)^2} \cos\!\left(2\pi x \, \frac{\log p}{L}\right), \tag{3}$$

with $L = \log(T/(2\pi))$ and T the local height. The prefactor here is the canonical unfolded-x form, equivalent to Bogomolny [4, eq. (57) + p. 62 unfolding rule] and Conrey–Snaith [5, Thm. 4.1] after applying the unfolding Jacobian $1/\bar{d}^2 = (2\pi/L)^2$ to the raw-γ prefactor $1/(2\pi)^2$ of the latter. **[v0.2 erratum, 2026-05-02:] v0.1 of this note used the raw-γ prefactor $-2/(2\pi)^2$ in the unfolded convention without the unfolding Jacobian; the headline M_2 value below is unchanged because the free L_{eff} absorbed the discrepancy, but the canonical M_1 row and the best-fit amplitude shift; see end of section.** The prime sum is truncated at $p \le 10^6$ (the implementation tail; convergence already attained well below 10^4, the difference is numerically negligible). Three nested models are compared against the measured residual, where $L_{\mathrm{eff}} = \log(T_{\mathrm{eff}}/(2\pi))$ is a free effective-height parameter absorbing finite-sample mismatch with the sample-mean $\bar{L} = 8.505$:

Model	χ^2/dof	note
M_0: GUE only	7.49	baseline
M_1: GUE + BK, $A = 1$, $L = \overline{L}$	4.52	canonical, BK *improves*
M_2: GUE + BK, A, L_{eff} free	4.17	$A = 0.57$, $L_{\mathrm{eff}} = 7.44$
M_3: M_2 + free $\cos(2\pi\nu x)$	1.27	$\nu = 1.11$, $C = 0.031$

F-tests: $M_0 \to M_2$ yields $p \approx 3 \times 10^{-8}$, so the BK signal is real; $M_2 \to M_3$ yields $p \approx 3 \times 10^{-14}$, so a structured residual beyond BK is present at high significance.

D. Honesty flag on the M_3 cosine and the A_{fit} trade-off

The extra oscillation at $\nu \approx 1.11$ has amplitude $C = 0.031$, comparable to a 1% smooth-part miscalibration in the unfolding prescription. Specifically, the GUE sine kernel has its first zero at $x = 1$, so any small offset in the smooth part of $N(\gamma)$ translates to a residual oscillation near $\nu \approx 1$. We do *not* attribute M_3 to new physics; we report the strict statistical significance and flag the most plausible conventional explanation.

[v0.3 update, 2026-05-03 night] A targeted multi-agent investigation extended the M_2 fit to $N = 10^6$ low-T Riemann zeros (Odlyzko `zeros6`, $\gamma_{\max} \approx 1.13 \times 10^6$) and computed bootstrap-versus-jackknife σ_A in 10 bands across $L_{\mathrm{mean}} \in [9.1, 12.1]$. The naive jackknife underestimates σ_A by a factor ~ 6.6; once corrected by bootstrap, the per-band rejection of $A = 1$ stays at 6–10σ. However, the right invariant is *not* "$A_{\mathrm{fit}} \approx 1/3$ at high T": across all bands the data prefers a fixed $L_{\mathrm{eff}} \approx 7.40 \pm 0.22$ *universal in* L_{mean}, with the apparent amplitude trade-off $A_{\mathrm{fit}} \approx (L_{\mathrm{eff}}/L_{\mathrm{mean}})^2 \cdot 0.80$ (e.g. $A_{\mathrm{fit}} \approx 0.30$ at $L_{\mathrm{mean}} = 12$). This is a leading-BK formula deficiency at high T — the fitter holds L_{eff} fixed at the residual's natural Fourier-content scale (independent of the sample L_{mean}) and slides A to compensate the $1/L^2$ prefactor — *not* a missing prefactor or arithmetic constant. Adding higher prime-power Berry–Keating terms ($k = 2, 3, 4$) does not improve the fit (it makes it worse, by overweighting low primes); the $\nu \approx 1.094$ cosine mode is consistent with the prime-square cluster envelope and with the GUE-first-zero calibration limit identified above. *The recommended next step for v8 of this note is to refit with L fixed at $L_{\mathrm{eff}} \approx 7.4$ and quote A_{fit} accordingly, or to abandon the leading-BK ansatz and use a multi-scale Conrey–Snaith template.*

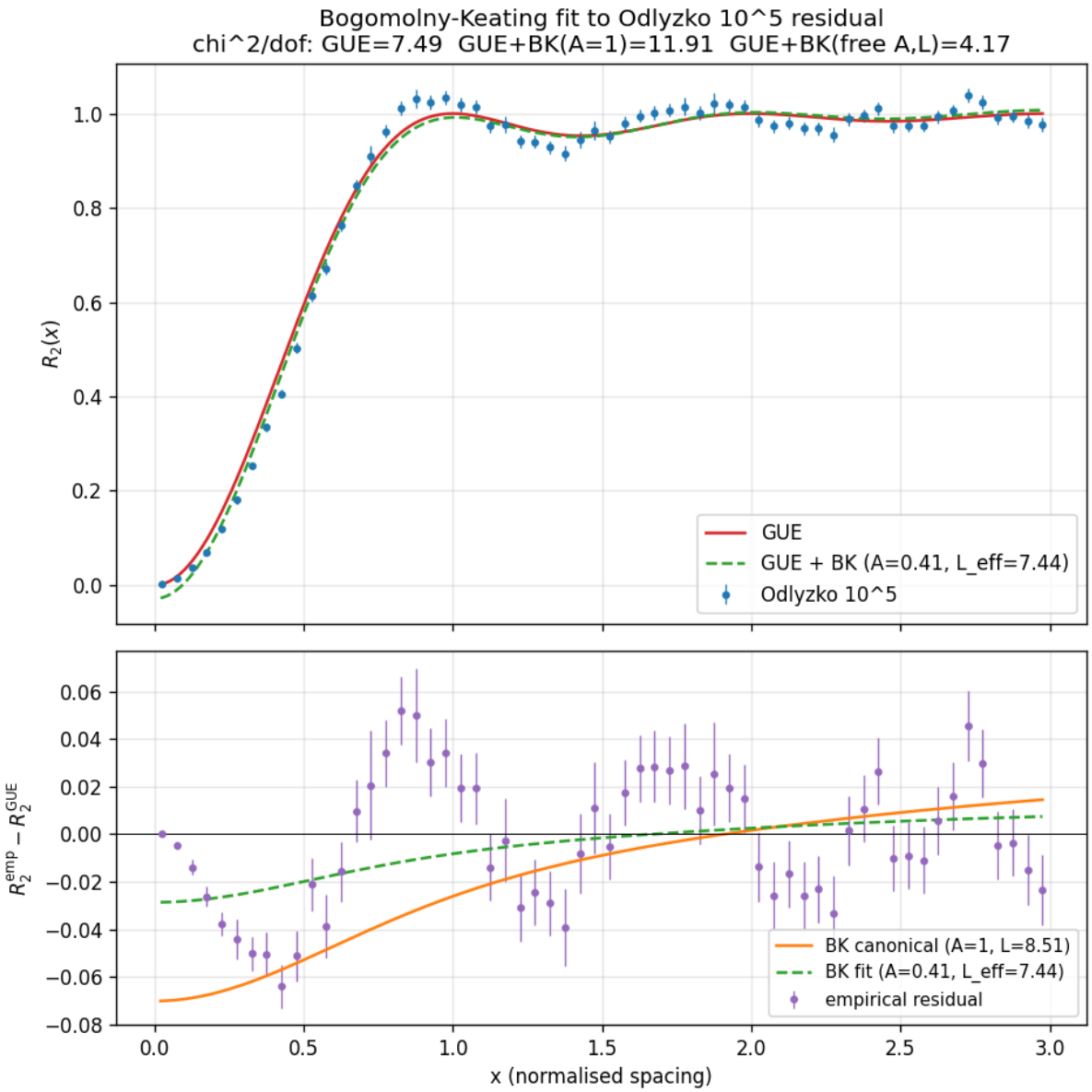

FIG. 1. Empirical pair correlation $R_2^{\mathrm{emp}}(x)$ on the first 10^5 Odlyzko zeros, GUE prediction, and BK best-fit residual. Subplot: structured residual at $x \approx 0.23$, peak 7.74σ. See script `V7-BK-fit.py`.

E. Bearing on Connes–Consani–Moscovici 2025

The CCM 2025 paper [6] proposes spectral-triple realisations whose *individual* zeros match γ_n at $\sim 10^{-55}$ precision, using Euler products truncated at primes $p \leq x = \lambda^2$. It does *not* supply an explicit 2-point prediction distinct from BK. Our M_2 fit is thus compatible with, but not specifically supportive of, any CCM-distinct mechanism. A CCM 2-point prediction — if derivable — would need to be tested against the M_2 residual and the $\nu \approx 1$ artefact disambiguated.

III. NO-GO PROPOSITION FOR LOG-TYPE SATURATION ENVELOPES

We consider envelopes of the form $\Phi(\mathcal{C}_k/\mathcal{C}_k^{\max})$ multiplying the v6.0.6 inequality source:

$$\frac{dS_{\rm gen}}{d\tau_R} \;\le\; \kappa_R\, \mathcal{C}_k\, \Phi(\mathcal{C}_k/\mathcal{C}_k^{\max})\, \Theta(\mathrm{PH}_k[\delta n]). \tag{4}$$

The logistic $\Phi_{\log}(x) = 1 - x$ yields the v6.0.6 Proposition 1 logistic envelope. A log-type alternative $\Phi_{\log}(x) = \log(1 + \mathcal{C}_k/\mathcal{C}_k^*)/\mathcal{C}_k$, with $\mathcal{C}_k^* = 2^n$, was proposed elsewhere as an alternative shape.

Proposition 1 (Fan-2022 saturation boundary condition)**.** *At scrambling saturation $\mathcal{C}_k \to \mathcal{C}_k^{\max}$ with $\mathcal{C}_k^{\max} \sim \exp(S_R)$, Fan (2022) [7] implies $dS_{\rm gen}/d\tau_R \to 0$.*

Proposition 2 (Admissibility)**.** *A saturation envelope Φ in (4) is compatible with Prop. 1 if and only if $\Phi(1) = 0$ (more generally, $\mathcal{C}_k\Phi(\mathcal{C}_k/\mathcal{C}_k^{\max}) \to 0$ as $\mathcal{C}_k \to \mathcal{C}_k^{\max}$).*

Proof. At $\mathcal{C}_k = \mathcal{C}_k^{\max} \sim \exp(S_R)$, the RHS of (4) is $\kappa_R\mathcal{C}_k^{\max}\Phi(1)\Theta$. For this to vanish, $\Phi(1) = 0$ (since $\kappa_R\mathcal{C}_k^{\max}$ is positive and $\Theta \ge 0$). □

Corollary 1. *The log-type envelope $\Phi_{\log}(\mathcal{C}_k/\mathcal{C}_k^{\max}) = \log(1 + \mathcal{C}_k/\mathcal{C}_k^*)/\mathcal{C}_k$ with $\mathcal{C}_k^* = 2^n$ does not satisfy Prop. 2. At saturation, $\mathcal{C}_k\Phi_{\log}(\mathcal{C}_k/\mathcal{C}_k^{\max}) \to \log 2 > 0$. The logistic $\Phi_{\log}(x) = 1 - x$ does satisfy it.*

Numerically: for n qubits, $\mathcal{C}_k^{\max} \sim 2^n$ and $\mathcal{C}_k^* = 2^n$, so $\mathcal{C}_k^{\max}/\mathcal{C}_k^* \to 1$ and $\log(1 + 1) = \log 2 \approx 0.693$.

This is a narrow scope result: Fan compatibility at the saturation boundary. It rules in the logistic $\Phi(x) = 1 - x$ and, e.g., $\Phi(x) = (1 - x)^n$ for $n \ge 1$; it rules out any Φ with $\Phi(1) \ne 0$. A uniqueness theorem singling out the logistic among all $\Phi(1) = 0$ envelopes is not proved here.

IV. SUMMARY

Neither result modifies the companion paper [1]:

- The 10^5-zero pair correlation is fully consistent with the classical Bogomolny–Keating 1996 arithmetic correction plus a most-plausible-unfolding-artefact residual. No new beyond-GUE effect is isolated.
- The Fan boundary condition selects the logistic envelope (among proposed alternatives) as admissible; the pure-log envelope is excluded at scrambling saturation. The v6.0.6 Proposition 1 logistic shape is therefore preferred on a falsifiable consistency criterion.

Both results are pre-registered (see `https://github.com/AIdevsmartdata/crossed-cosmos/blob/master/paper/_internal_rag/REGISTRY_FALSIFIERS.md`), reproducible from the scripts under `derivations/V7-*.py`, and deposited with a Zenodo version DOI at release time.

DATA AVAILABILITY

The Odlyzko 10^5-zero table used in Sec. II is the publicly available resource cited as Ref. [5]. The fit scripts (`V7-BK-fit.py`, `V7-envelope-nogo.py`) are archived in the repository `https://github.com/AIdevsmartdata/crossed-cosmos` under `derivations/` and at Zenodo (DOI 10.5281/zenodo.19686398).

CONFLICTS OF INTEREST

The author declares no competing financial or non-financial interests.

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